MIL-R-8903A (ASG)
14 FEBRUARY 1966
Superseding
MIL-R-8903(ASG)
5 April 1962

MILITARY SPECIFICATION

RESISTOR, VARIABLE, LINEAR PRECISION, TAPPABLE

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers single-section, precision, variable, wire-wound resistors, capable of producing an output voltage (in terms of percent of applied voltage) proportional to angle of shaft rotation. The resistors can be tapped by press fitting a particular pressure-type tap to a tap ring within the resistor. The resistors can be ganged by means of available mounting hardware consisting of a clamping ring and drive arm as shown on figure 1 (see 3.4). The resistors are for use in training equipment only.

1.2 Classification.- Resistor type designations shall be in the following form, as specified (see 6.2):

Type designation form - single-section resistors

Style	Terminal board	Resistance
RRL16	A	103

1.2.1 Style.- The style is identified by the three-letter symbol "RRL" followed by a two-digit number; the letters identify tappable, precision, linear, variable resistors, and the number identifies the physical size incrementally in 1/8-inch increments.

1.2.2 Terminal board.- Terminal boards shall be of two types, the letter A designating a 110-degree arc board, and the letter B designating 180-degree arc board (see 3.4.9).

1.2.3 Resistance.- The nominal total resistance value expressed in ohms is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

FSC 5905

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-200/3	Aluminum Alloy Bar, Rod, Shapes, and Tube Extruded 2024
QQ-S-763	Steel Bars, Shapes, and Forgings - Corrosion Resisting
QQ-S-766	Steel Plates, Sheets, and Strip - Corrosion Resisting
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-T-60	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-S-6872	Soldering Process, General Specification for
MIL-T-23099	Tapping Kit for Servo-Mount Potentiometers, Two-Inch Diameter

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-440	Soldering Techniques for Standard Type Solder Terminals
MS28902	Contact, Electrical, Leaf Tap for Use with MIL-R-8903 Resistor
MS28927	Clamp, Rim, Clenching
MS28928	Coupling, Insulated, Bushing Shaft, for MIL-R-8903 Resistor
MS28931	Cap, Protective, Dust and Moisture Seal for MIL-R-8903 Resistor

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Defense Supply Agency

Handbook H 4-1	Federal Supply Code for Manufacturers Part 1 - Name to Code

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National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies of above publications should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Qualification.- The resistor furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Definitions.- For definitions of terms used in this specification, see 6.4.

3.3 Material.- The material for each part shall conform to applicable specifications and as specified herein. However, when a definite material is not specified, a material shall be used which will enable the resistors to perform satisfactorily when subjected to the performance tests specified herein. The resistor shall have a shelf life of not less than 3 years.

3.3.1 Plastic material.- Plastic laminates containing a cotton-fabric base, or plastic molding compounds containing a cotton or wood flour filler, shall not be used.

3.3.2 Ferrous metals.- The use of ferrous metals, with the exception of corrosion-resistant steel and the resistance-element material is prohibited. Corrosion-resistant steel used shall be in accordance with QQ-S-763 and QQ-S-766.

3.4 Design and construction.- The resistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4.1 Lateral runout.- When the resistor is tested as specifiied in 4.7.1, the lateral runout (see figure 2) shall not exceed 0.001 total indicator reading (tir).

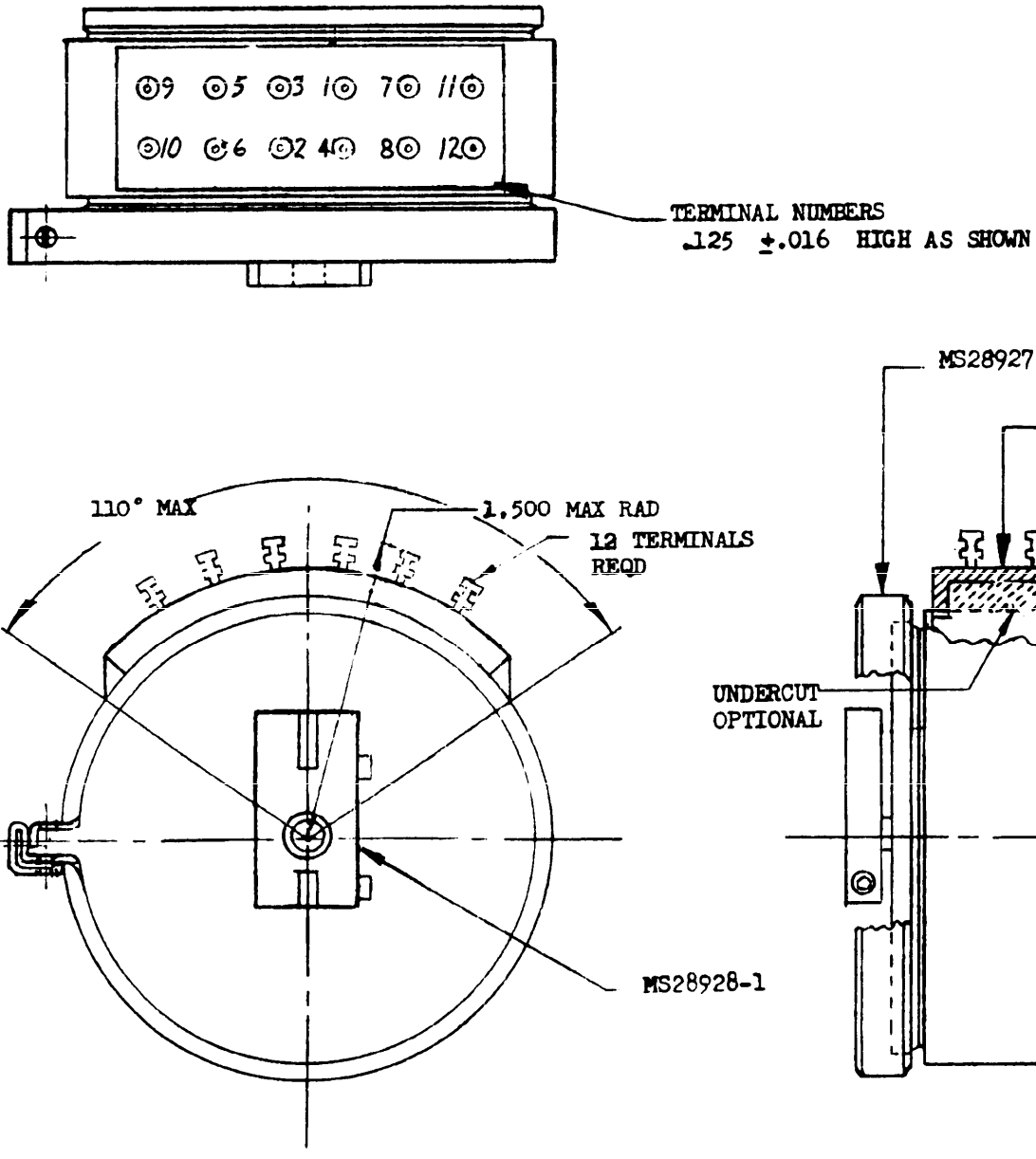
3.4.2 Pilot-surface runout.- When the resistor is tested as specified in 4.7.2, the pilot surface runout (see figure 3) shall not exceed 0.001 in tir.

3.4.3 End play.- When the resistor is tested as specified in 4.7.3, the end play (see figure 4) shall not exceed 0.006 inch.

3.4.4 Electrical travel.- The electrical travel shall be 320 degrees ± 1 degree.

3.4.5 Overtravels.- The overtravels at each end of the resistance element shall be a minimum of 10 degrees each and shall be made of nominal zero resistance material.

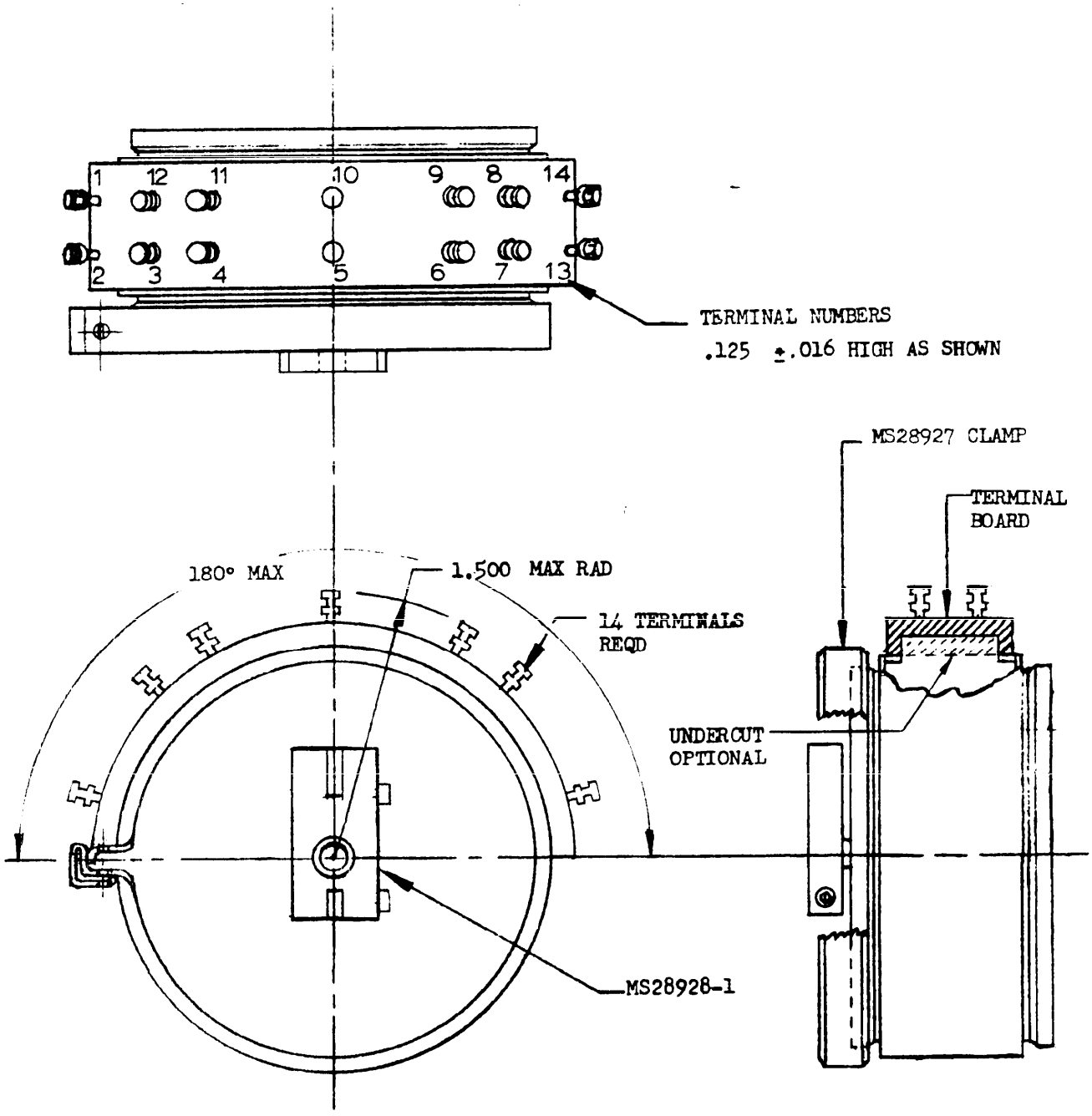
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DIMENSIONS IN INCHES.

FIGURE 1 (Sheet 1 of 4). Resistor, variable

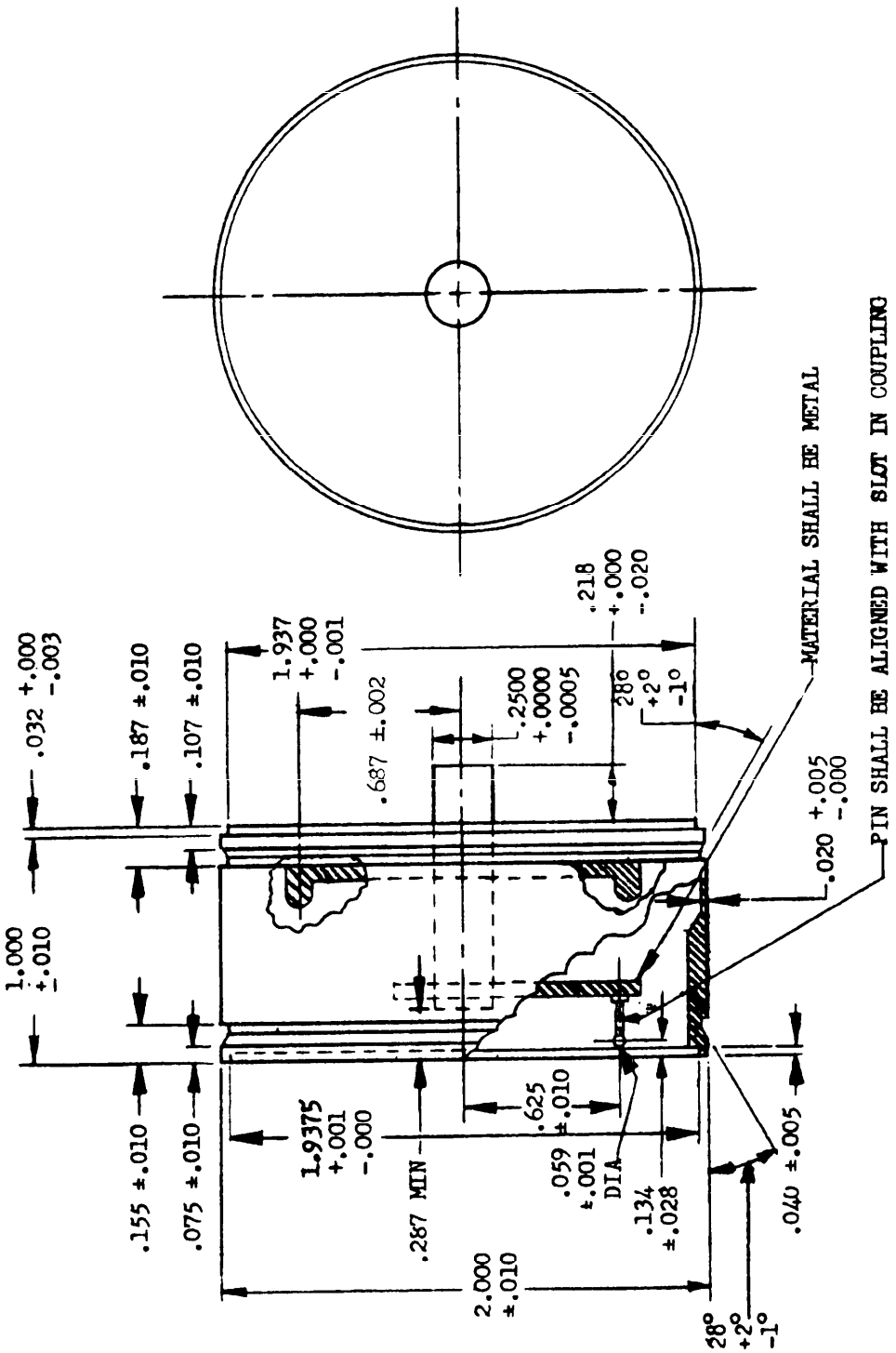
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DIMENSIONS IN INCHES.

FIGURE 1 (Sheet 2 of 4). Resistor, variable

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DIMENSIONS IN INCHES.

FIGURE 1 (Sheet 3 of 4). Resistor, variable

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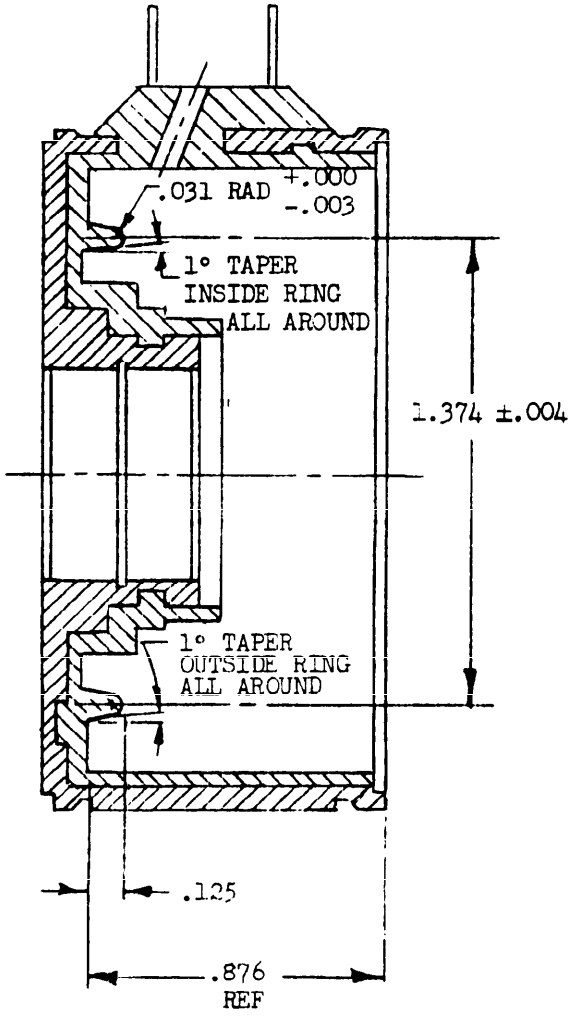


FIGURE 1 (Sheet 4 of 4). Resistor variable

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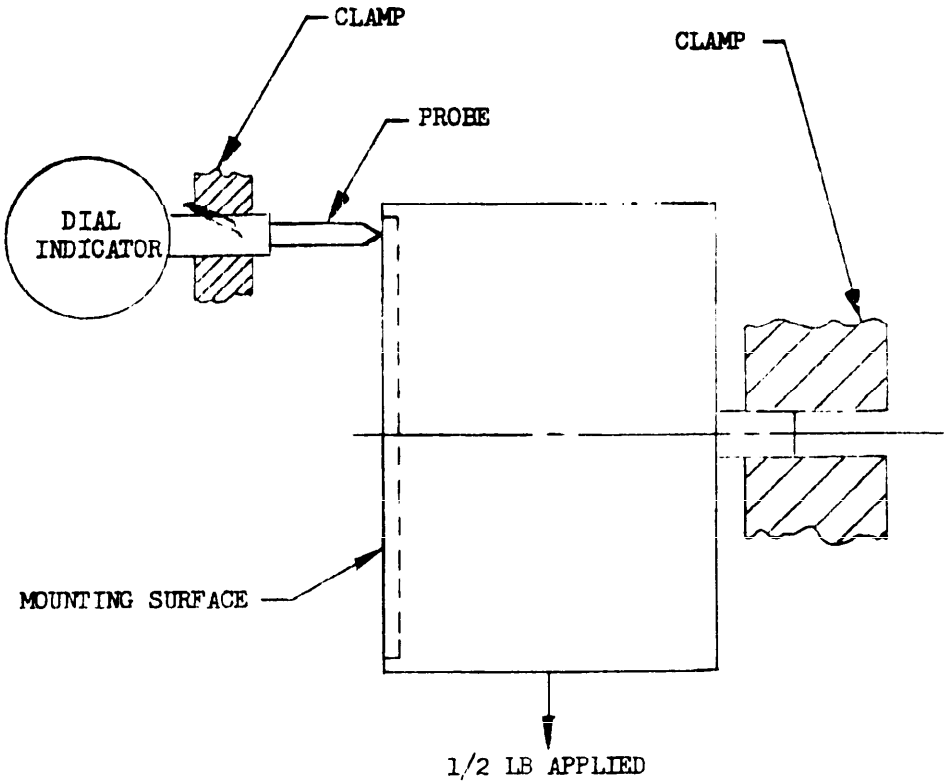


FIGURE 2. Measurement of lateral runout

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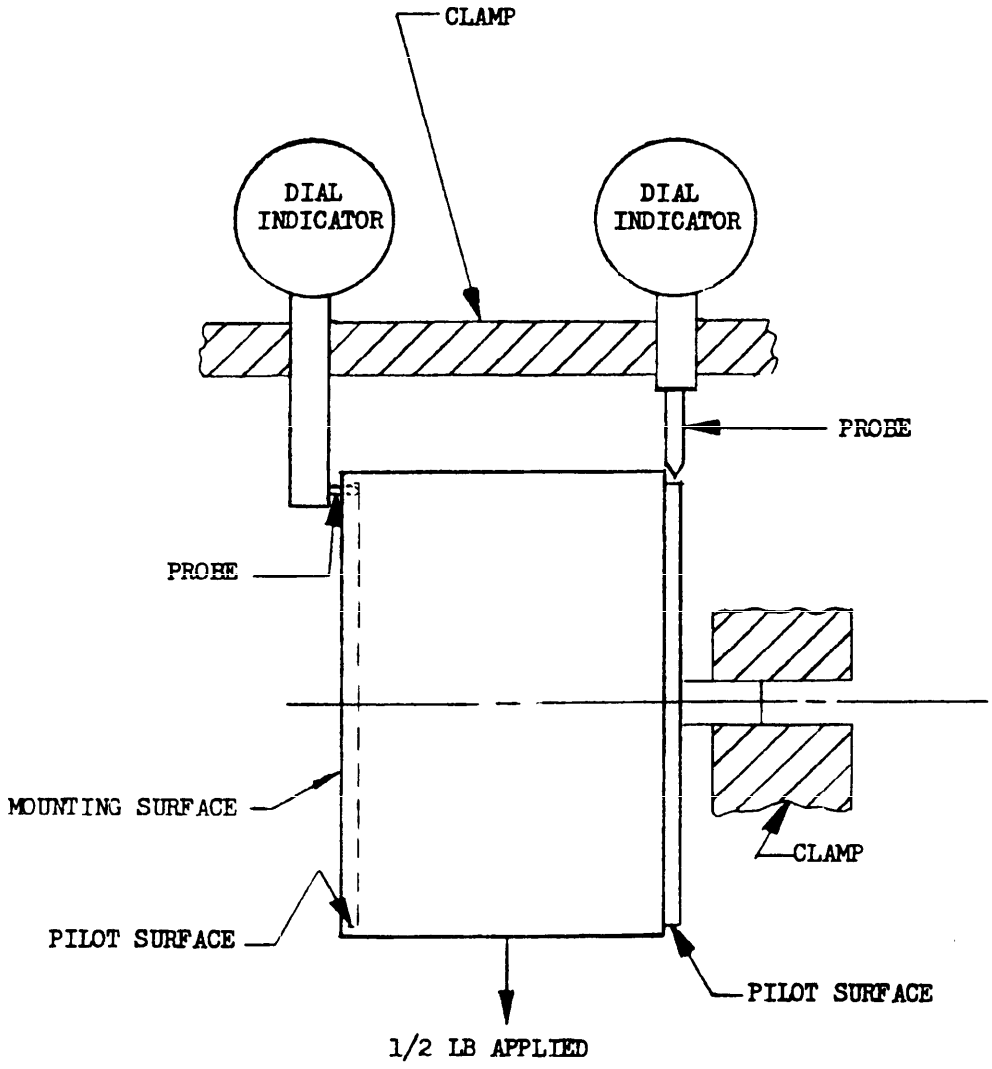


FIGURE 3. Measurement of pilot surface runout

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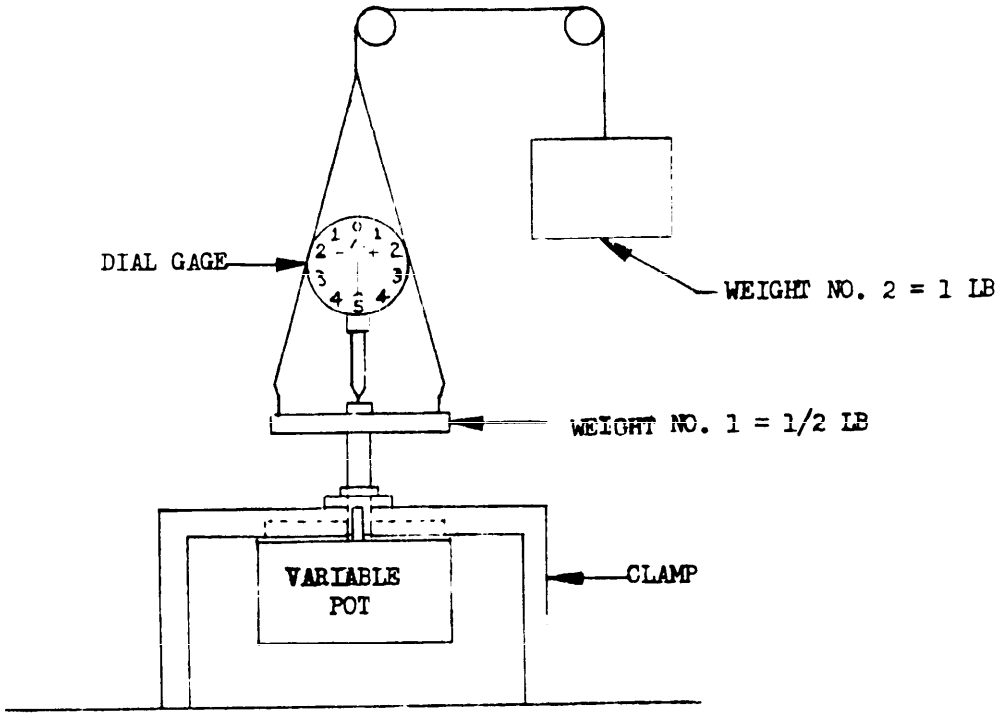


FIGURE 4. Measurement of end play

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3.4.6 Contacts.- The variable electrical connection to the resistance element shall be made by a wiper and slip ring coupling.

3.4.6.1 Wiper.- The moving contact (wiper) shall have continuous electrical contact throughout the continuity travel and shall be insulated from the shaft and housing.

3.4.6.2 Slip ring coupling.- The slip ring coupling shall consist of a fixed metal ring and a moving spring contact assembly. The spring assembly shall consist of dual contacts to bear on the sides or edges of the fixed ring. The slip ring coupling shall be insulated from the shaft and housing.

3.4.7 Housing.- The resistor housing shall be made of aluminum alloy conforming to QQ-A-200/3 and shall be of one-piece construction in accordance with the dimensions shown on figure 1.

3.4.8 Taps.- When required, the resistor element shall be tapped by means of taps which depend on a pressure contact with the resistance winding for electrical continuity and are retained in place by press fitting to a tap ring which forms a part of the resistor assembly. The dimensions for the tap shall be as shown on MS28902. Tapping may be accomplished by use of a tapping kit in accordance with MIL-T-23099.

3.4.8.1 Tap location.- When required, taps shall be capable of installation with a minimum separation of 10 degrees and within $\pm$0.6 degree of the specified angular location.

3.4.9 Terminal board.- The terminal board assembly shall consist of a curved base with terminals as shown on figure 1. When mounted as shown on figure 1, the base of the terminal board shall conform to an arc on the periphery of the housing subtended by an angle of 110 or 180 degrees from the center of the resistor assembly. Terminals 1 and 3 shall be as close to the ends of the resistance element as possible.

3.4.10 Designations.- Terminals shall be connected and numbered with the number 2 terminal as the wiper terminal, number 1 terminal as the extreme clockwise terminal, and number 3 terminal as the extreme counterclockwise terminal, looking at the open end of the unit.

3.4.11 Coupling.- The resistor shall be provided with a coupling in accordance with MS28928.

3.4.12 Screw threads.- Screw threads for all screws, nuts, and other threaded parts shall be of American National Form in accordance with Handbook H28 and shall be class 2 fit.

3.4.12.1 Securing of threaded parts.- Threaded mounting hardware shall be properly secured to withstand the vibration test specified in 4.7.17 without loosening.

3.5 Torque, initial.- The torque required to rotate the contact arm shall be not more than 0.6 ounce-inch (see 4.7.4).

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3.6 Power rating.- The resistor shall have a power rating of four watts based on continuous full load operation at an ambient temperature of 40° C. For operation above 40° C, the resistor shall be derated proportionally to zero wattage at 105° C. The power rating shall be met when the resistor is mounted in accordance with 4.7.15.1.1. If mounted with the open end of the resistor exposed, a cover conforming to MS28931 shall be used.

3.7 Direct current (dc) resistance.-

3.7.1 Total resistance.- When measured as specified in 4.7.5.1, the total dc resistance shall be equal to the nominal resistance value ±1 percent. The standard nominal values shall be 100, 200, and 500 ohms, 1, 2, 2.5, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 50, 75, 100, 150, and 200 kilohms.

3.7.2 End resistance.- When measured as specified in 4.7.6, the end resistance shall not exceed 1 ohm.

3.7.3 Tapped resistance.- When measured as specified in 4.7.7, the tapped resistance shall be equal to the nominal tapped resistance value ±1 percent.

3.8 Insulation resistance.-

3.8.1 Initial.- When the resistor is tested as specified in 4.7.8, the insulation resistance shall be not less than 1,000 megohms.

3.8.2 Degradation.- When the resistor is tested as specified in 4.7.8, the insulation resistance shall be not less than 100 megohms.

3.9 Dielectric withstanding voltage.- When the resistor is tested as specified in 4.7.9, there shall be no damage, arcing, or breakdown. The leakage current shall not exceed 1 milliampere (ma).

3.10 Linearity.- Variable resistor linearity is the maximum percent deviation of the resistor performance curve from the desired linear function of the resistance winding as determined by either percent voltage or percent resistance versus percent shaft rotation. The percent deviation for any angular position of the wiper is the difference between the two curves at that point.

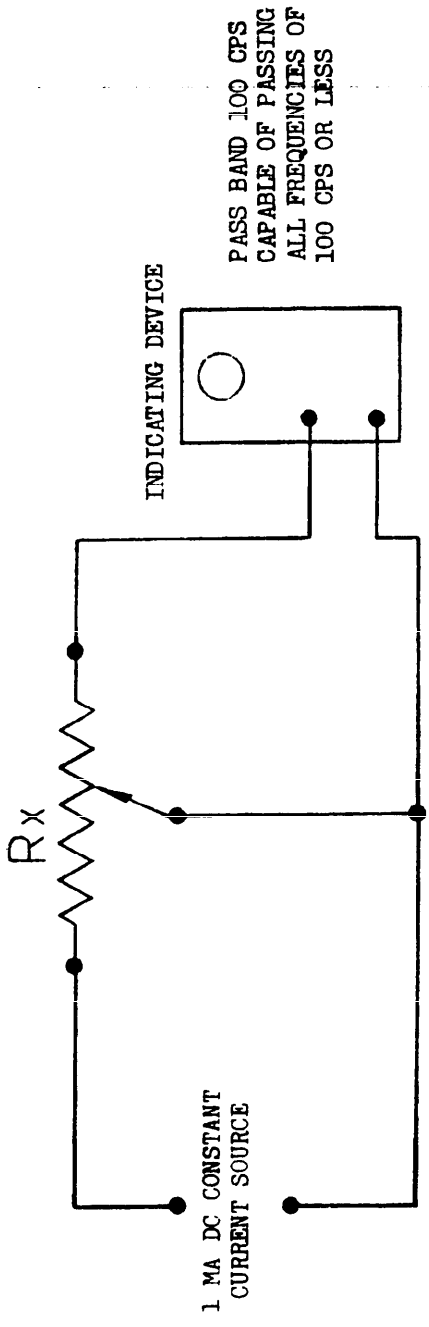
3.10.1 Linearity tolerance.- When the resistor is tested as specified in 4.7.10, the terminal linearity tolerance shall be not more than ±0.5 percent of the total applied voltage.

3.11 Quadrature voltage.- When tested as specified in 4.7.12, the quadrature voltage ratio shall not exceed 2 millivolts/volt applied.

3.12 Noise (equivalent noise resistance).- When the resistor is tested as specified in 4.7.11, the equivalent peak-to-peak noise resistance shall not exceed 140 ohms (see figure 5).

3.13 Temperature cycling.- When the resistor is tested as specified in 4.7.13, there shall be no mechanical damage and the total resistance change shall not exceed 1 per cent of the total resistance value (see 3.7.1).

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3.14 Resistance-temperature characteristic.- When the resistor is tested as specified in 4.7.14, the resistance-temperature characteristic, referred to ambient temperature of 40° C, shall not exceed ±0.003 percent per degree centigrade.

3.15 Rotational life.- When the resistor is tested as specified in 4.7.15.1, there shall be no mechanical damage or opening of the winding, and the total-resistance change shall not exceed 5 percent between the total resistance value (see 3.7.1) and any of the succeeding measurements. Linearity shall be within ±1.5 percent, noise 2,000 ohms maximum, and torque 1 ounce inch maximum.

3.16 Load life.- When the resistor is tested as specified in 4.7.15.2, there shall be no mechanical damage or opening of the winding, and the total change shall not exceed 1 percent between the total resistance value and any of the succeeding measurements. Linearity shall be within ±1.5 percent and noise 2,000 ohms maximum.

3.17 Low-temperature exposure.- When the resistor is tested as specified in 4.7.16, there shall be no mechanical damage, and the total resistance change shall not exceed 1 percent of the total resistance value. The change in tapped resistance shall not exceed 1 percent of the tapped resistance value (see 3.7.1 and 3.7.3).

3.18 Vibration, high frequency.- When the resistor is tested as specified in 4.7.17, there shall be no mechanical or electrical damage or momentary discontinuity. The total resistance change shall not exceed 1 percent of the total resistance value. The change in tapped resistance shall not exceed 1 percent of the tapped resistance value (see 3.7.1 and 3.7.3).

3.19 Shock.- When the resistor is tested as specified in 4.7.20, there shall be no mechanical or electrical damage or momentary discontinuity. The total resistance change shall not exceed 1 percent of the total resistance value. The change in tapped resistance shall not exceed 1 percent of the tapped resistance value (see 3.7.1 and 3.7.3).

3.20 Salt spray (corrosion).- When the resistor is tested as specified in 4.7.18, there shall be no corrosion, and the resistor shall remain mechanically operational and electrically continuous.

3.21 Humidity cycling.- When the resistor is tested as specified in 4.7.19, there shall be no mechanical damage, and the resistance change shall not exceed 5 percent of the total resistance value (see 3.7.1). Linearity shall be within ±1.5 percent, noise 2,000 ohms maximum, and torque 1 ounce inch maximum.

3.22 Cover.- When specified, a cover conforming to MS28931 shall be provided for the open end of the resistor (see 6.2).

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3.23 Identification of product.- The resistor shall be marked in accordance with MIL-STD-130. Marking shall remain legible at the end of all tests and shall include the following information:

Type designation
Federal Stock Number (when required by ordering data)
Federal Supply Code (in accordance with Handbook H 4-1)
Date (four-digit number; first two indicate month, second two indicate year)

3.24 Workmanship.- The resistor shall be meanufactured and processed in a careful and workmanlike manner in accordance with good design and sound practice, and shall conform to all applicable requirements of this specification.

3.24.1 Soldering.- Soldering shall be in accordance with MIL-STD-440 and MIL-S-6872.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The examination and testing of the resistors shall be classified as follows:

(a) Qualification inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Qualification inspection.-

4.3.1 Sampling instructions.- Thirty-six specimens, comprising a sample of the resistors, shall be submitted for qualification inspection. Eighteen of the specimens shall be of 100 ohms resistance and 18 of 100 kilohms. All samples shall be tapped at 80 degrees, 160 degrees, and 240 degrees.

4.3.2 Inspection routine.- The two values of specimens will be subjected to the examinations and tests specified in table I in the order shown. Eighteen specimens of each value shall be subjected to the examinations and tests of group 1. After examinations and tests of group 1, the 18 specimens of each value shall be divided into the remaining groups shown in table I, and subjected to the tests for their particular group.

4.3.3 Inspections.- The qualification inspections shall consist of all the examinations and tests specified in table I.

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TABLE I. Qualification inspection

Group and test	Requirement paragraph	Test method paragraph
Group 1: 18 specimens (of each value)		
Visual and mechanical examination	3.3 to 3.4 incl., 3.4.4 to 3.4.12 incl., and 3.23 to 3.24 incl.	4.6
Lateral runout	3.4.1	4.7.1
Pilot-surface runout	3.4.2	4.7.2
End play	3.4.3	4.7.3
Torque, initial	3.5	4.7.4
Total resistance	3.7.1	4.7.5.1
End resistance	3.7.2	4.7,6
Tapped resistance	3.7.3	4.7.7
Insulation resistance - Initial	3.8.1	4.7.8
Dielectric withstanding voltage	3.9	4.7.9
Linearity tolerance	3.10.1	4.7.10
Quadrature voltage	3.11	4.7.12
Noise (equivalent noise resistance)	3.12	4.7.11
Group 2: 6 specimens (of each value)		
Resistance-temperature characteristics	3.14	4.7.14
Rotational life	3.15	4.7.15.1
Linearity tolerance	3.10.1	4.7.10
Noise (equivalent noise resistance)	3.12	4.7.11
Insulation resistance - Degradation	3.8.2	4.7.8
Dielectric withstanding voltage	3.9	4.7.9
Torque, initial	3.5	4.7.4
Temperature cycling	3.13	4.7.13
Group 3: 3 specimens (of each value)		
Low-temperature exposure	3.17	4.7.16
Linearity tolerance	3.10.1	4.7.10
Tapped resistance	3.7.3	4.7.7
Noise (equivalent noise resistance)	3.12	4.7.11
Torque, initial	3.5	4.7.4
Load	3.16	4.7.15.2
Linearity tolerance	3.10.1	4.7.10
Tapped resistance	3.7.3	4.7.7
Noise (equivalent noise resistance)	3.12	4.7.11
Insulation resistance - Degradation	3.8.2	4.7.8
Dielectric withstanding voltage	3.9	4.7.9
Torque, initial	3.5	4.7.4
Temperature cycling	3.13	4.7.13

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TABLE I. Qualification inspection (continued)

Group and test	Requirement paragraph	Test method paragraph
Group 4: 3 specimens (of each value)		
Shock, medium impact	3.19	4.7.20
Vibration, high frequency	3.18	4.7.17
Linearity tolerance	3.10.1	4.7.10
Tapped resistance	3.7.3	4.7.7
Noise (equivalent noise resistance)	3.12	4.7.11
End play	3.4.3	4.7.3
Torque, initial	3.5	4.7.4
Salt spray (corrosion)	3.20	4.7.18
Temperature cycling	3.13	4.7.13
Group 5: 6 specimens (of each value)		
Humidity cycling	3.21	4.7.19
Linearity tolerance 1/	3.10.1	4.7.10
Noise (equivalent noise resistance) 1/	3.12	4.7.11
Torque, initial 1/	3.5	4.7.4
Temperature cycling	3.13	4.7.13

1/ These tests will be performed immediately following the humidity-cycling test (4.7.19).

4.3.4 Defectives.- The 18 specimens of each value shall be evaluated separately as specified. Failure of a resistor in one or more tests of a group will be charged as a single defective. No defective will be allowed for groups 1, 3, and 4 combined. Two defectives will be allowed for groups 2 and 5 combined. Defectives in excess of those allowed will be cause for refusal to grant qualification.

4.4 Quality conformance inspection.- Quality conformance inspection shall consist of sampling plan A and B inspections.

4.4.1 Inspection lot.- An inspection lot shall be as specified in MIL-STD-105.

4.4.2 Resubmitted lots.- If an inspection lot is rejected, the manufacturer may rework the lot or screen out defectives and resubmit it for product acceptance inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lots shall be inspected using tightened inspection.

4.4.3 Sampling plan A inspection.- Sampling plan A inspection shall consist of 100 percent inspection in accordance with table II. The Acceptance Quality Levels (AQL) shall be 1.0 (percent defective) for minor defects and 4.0 (percent defective) for major defects (see MIL-STD-105).

4.4.3.1 Disposition of sample units.- Sample units which have been subjected to sampling plan A inspection may be delivered on the contract or order if the lot is accepted.

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TABLE II. Sampling plan A inspections

Examination or test	Defect 1/ classification	Requirement paragraph	Test method paragraph
Total resistance	Major	3.7.1	4.7.5.1
Insulation resistance - initial	Major	3.8.1	4.7.8
Dielectric withstanding voltage	Major	3.9	4.7.9
Linearity tolerance	Major	3.10.1	4.7.10
Quadrature voltage	Major	3.11	4.7.12
Noise (equivalent noise resistance)	Major	3.12	4.7.11

1/ Major and minor defects shall be as defined in MIL-STD-105.

4.4.4 Sampling plan B inspection.- Sampling plan B inspection shall consist of those inspections specified in table III. Statistical sampling and inspection shall be in accordance with MIL-STD-105. The inspections shall be performed on sample units which have been subjected to and have passed sampling plan A inspection.

4.4.4.1 Sampling procedure.- The sampling procedure shall be in accordance with MIL-STD-105 for subgroups 1, 2, and 3. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The AQL shall be 4.0 (percent defective) and the inspection level shall be S-4 for normal and tightened inspection and S-3 for reduced inspection.

4.4.4.2 Disposition of sample units.- Sample units which have been subjected to subgroups 2 and 3 tests shall not be delivered on the contract or order. Sample units which have been subjected to subgroup 1 inspection may be delivered on the contract or order if the lot is accepted.

4.5 Test conditions and methods.- Unless otherwise specified herein, all tests shall be made at room ambient temperature, pressure, and humidity. All electrical tests shall be made at rated voltage unless otherwise specified.

4.5.1 Methods.- The theoretically calculated rated voltage or the voltage equivalent to the power rating of 4 watts shall be determined from the following formula:

$$E = (PR)\ 1/2 = \sqrt{PR}$$

Where:

E = rated dc or root mean square (rms) voltage
P = power rating
R = nominal resistance

Example $E = (PR)^{1/2} = (4R)^{1/2} = 2R^{1/2}$

or $E = \sqrt{PR} = \sqrt{4R} = 2\sqrt{R}$

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TABLE III. Sampling plan B inspections 1/

Examination or test	Defect classification 2/	Requirement paragraph	Test method paragraph
Subgroup 1			
Visual and mechanical examination	Major	3.3 to 3.4 incl.	4.6
Lateral runout	Major	3.4.1	4.7.1
Pilot-surface runout	Major	3.4.2	4.7.2
End play	Major	3.4.3	4.7.3
Torque, initial	Major	3.5	4.7.4
End resistance	Major	3.7.2	4.7.6
Identification of product	Major	3.23	---
Workmanship	Major or minor	3.24 to 3.24.1 incl.	---
Subgroup 2			
Temperature cycling	Major	3.13	4.7.13
Rotation 3/	Major	3.15	4.7.15
Linearity tolerance	Major	3.10.1	4.7.10
Noise (equivalent noise resistance)	Major	3.12	4.7.11
Insulation resistance - initial	Major	3.8.1	4.7.8
Dielectric withstanding voltage	Major	3.9	4.7.9
Torque, initial	Major	3.5	4.7.4
Subgroup 3			
Temperature cycling	Major	3.13	4.7.13
Vibration, high frequency	Major	3.18	4.7.17
Linearity tolerance	Major	3.10.1	4.7.10
Noise (equivalent noise resistance)	Major	3.12	4.7.11
End play	Major	3.4.3	4.7.3
Torque, initial	Major	3.5	4.7.4

1/ Failure of a resistor in one or more tests of a subgroup will be charged as a single defective.

2/ Major and minor defects shall be as defined in MIL-STD-105.

3/ This is the first part of the rotational life test only, and shall be performed on an accelerated basis in accordance with 4.7.15.1.

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4.6 Visual and mechanical examination.- Resistors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.7 Test methods.-

4.7.1 Lateral runout.- The operating shaft shall be clamped horizontally in a suitable device, and a dial indicator positioned so that the probe will contact the mounting surface of the resistor at the maximum possible distance from the operating shaft. The lateral runout shall be determined from the maximum positive and negative deviations added together without regard to algebraic signs (tir) (see 3.4.1).

4.7.2 Pilot-surface runout.- The operating shaft shall be clamped in a suitable device, and a dial indicator so positioned that the probe will contact the periphery of the pilot surface. The resistor body shall be rotated through 360 degrees. The runout of the pilot surface shall be the maximum positive and negative deviations added together without regard to algebraic signs (tir) (see 3.4.2).

4.7.3 End play.- The resistor body shall be clamped in a suitable device and the axial movement of the shaft measured with a dial gage with a 1/2-pound axial load on the shaft applied alternately in each direction. The end play shall be the total indicator reading (see 3.4.3).

4.7.4 Torque, initial.- The torque required to rotate the contact arm on the resistance element shall be determined throughout the entire range of mechanical travel by a torque tester, the torque-wrench method, or by any other method satisfactory to the procuring activity (see 3.5).

4.7.5 Dc resistance:- Resistors shall be tested in accordance with Method 303 of MIL-STD-202. The following details shall apply:

(a) The same bridge shall be used for all resistance measurements in any one test, but not necessarily for all the tests.
(b) Test voltage: The test voltage shall be in accordance with table IV.

TABLE IV. Resistance test voltage (dc)

Total resistance, nominal	Maximum test voltage 1/
Ohms	Volts
100 to 1,000,incl.	3.0
Over 1,000 to 10,000,incl.	10
Over 10,000 to 200,000,incl.	30

1/ The voltage used for the initial test shall be used for all subsequent tests.

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4.7.5.1 Total (initial and subsequent).- Total resistance shall be measured as specified in 4.7.5. Before each measurement of total resistance, the operating shaft shall be rotated 10 times through the electrical travel. Total resistance shall be measured between the resistance-element end terminals, and the contact arm positioned off the effective winding. The positioning of the contact arm and terminals shall be the same for all subsequent measurements of the total resistance on the same specimen (see 3.7.1).

4.7.6 End resistance.- Resistance shall be measured as specified in 4.7.5. The contact arm shall be positioned at its extreme counterclockwise limit of effective electrical travel. With the contact arm in this position, the resistance between terminal No. 1 and the contact-arm terminal shall be measured. The contact arm shall then be positioned at its extreme clockwise limit of effective electrical travel. With the contact arm in this position, the resistance between terminal No. 3 and the contact-arm terminal shall be measured (see 3.7.2).

4.7.7 Tapped resistance.- The tapped resistance shall be measured as specified in 4.7.5. Tapped resistance shall be measured between the taps and a resistance-element end terminal. The end terminal used for the initial test shall be used for all subsequent tests. During this test, precaution shall be taken to insure that the maximum current of the resistance element is not exceeded (see 3.7.3).

4.7.8 Insulation resistance (see 3.8).- The resistor shall be tested in accordance with Method 302 of MIL-STD-202. The following details shall apply:

(a) Test-condition, as specified in test condition letter B.
(b) Special preparations and conditions:
    (1) Room conditions: This test shall be conducted at a temperature of 23° ±1° C, a relative humidity of 50 ±2 percent, and an atmospheric pressure of 30 inches of mercury.
    (2) Mounting: For qualification inspection, the resistor shall be mounted on a metal plate of sufficient size to extend beyond the resistor extremities. For all other testing, the use of the metal plate is optional. The metal plate and the shaft shall make electrical contact with each other. All terminals of the resistor shall be tied together.
(c) The points of measurement shall be between the terminals connected together and the shaft.

4.7.9 Dielectric withstanding voltage (see 3.9).- The resistor shall be tested in accordance with Method 301 of MIL-STD-202. The following details shall apply:

(a) Special preparations and conditions, as specified in 4.7.8(b)(2).
(b) The magnitude of test voltage shall be 1,000 volts rms.
(c) Nature of potential: From an alternating-current (ac) supply at commercial-line frequency and waveform. This potential shall be applied with the operating shaft traversing the electrical travel in 1 minute.
(d) Points of application of test voltage, as specified in 4.7.8(c).

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the oscilloscope screen.

4.7.12 Quadrature voltage (see 3.11).- The quadrature voltage ratio shall be measured with the measuring circuit shown on figure 6 in accordance with the following procedure:

(a) Set the pot under test to approximately 25-percent rotation. Adjust the ratiotran to obtain a null signal (En) on the VTVM. Record the value of this voltage. This is the quadrature component of the potentiometer output signal.
(b) Repeat the measurement approximately 50 percent and 75 percent rotation.
(c) Measure the voltage Et applied to the pot under test and compute the quadrature voltage ratio (En/Et) for each value of En.

4.7.13 Temperature cycling (see 3.13).- The resistor shall be tested in accordance with Method 102 of MIL-STD-202. The following details shall apply:

(a) Special mounting: The resistor shall be mounted by its normal mounting means in such manner that there is at least 1 inch of free air space around the resistor. The mounting shall be so positioned with respect to the air stream that it offers the least obstruction to the flow of air across and around the resistor.
(b) Measurements before cycling: Not applicable.
(c) Test conditions, as specified in test condition letter D.
(d) Measurements after cycling: Total resistance shall be measured as specified in 4.7.5.1, and the resistor shall be examined for evidence of mechanical damage.

(e) Examinations and measurements: During the test, the leakage current shall be measured. At the conclusion of the test, the resistor shall be examined for evidence of damage, arcing, and breakdown.

4.7.10 Linearity tolerance.- Linearity measurements shall be performed at intervals of not greater than 5 percent of the electrical travel. The resistor shall be tested without load. Measurements shall be performed by using a dc potential of low enough value to insure that its application for the time necessary to obtain readings does not appreciably affect the measured values. The method employed for the measurement of linearity shall be such that the combined inherent errors of the measuring instruments shall not exceed 0.1 percent (see 3.10).

4.7.11 Noise (equivalent noise resistance (see 3.12).- Equivalent noise resistance shall be measured with the measuring circuit shown on figure 5, its equivalent, or a ponogometer. During this test, the operating shaft shall be rotated in both directions at an angular rate not exceeding four revolutions per minute (rpm). The wiper arm shall be rotated through 10 complete revolutions prior to conducting the noise test. The equivalent resistance shall be calculated, using the following formula:

$$\text{Noise} = \frac{\text{Ego}}{0.001}$$

Where:

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TEST PROCEDURE:

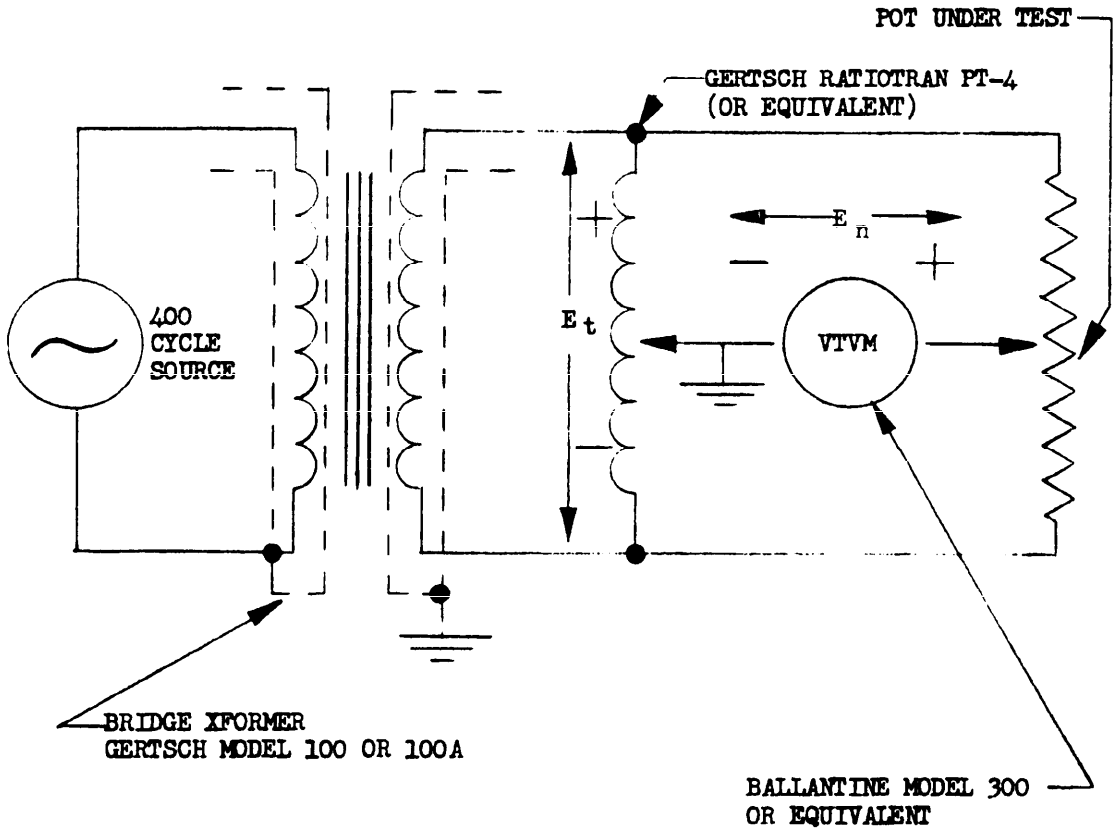


FIGURE 6. Quadrature voltage measuring circuit

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4.7.14 Resistance-temperature characteristic (see 3.14).- The resistor shall be tested in accordance with Method 304 of MIL-STD-202. The following details shall apply:

(a) Test temperatures: As specified in Method 304, except the highest temperature shall be 105° C.
(b) Measurements at end of each period: Total resistance shall be measured as specified in 4.7.5.1 at the temperature maintained during this period.

4.7.15 Rotational load life (see 3.15).-

4.7.15.1 Rotation.-

4.7.15.1.1 Mounting.- The resistor shall be centrally mounted by its normal mounting means on a 4-inch-square, 0.250-inch-thick, aluminum-alloy panel in accordance with QQ-A-200/3.

4.7.15.1.2 Test conditions.- This test shall be conducted in an ambient temperature of 80° ±2° C. The chamber dimensions shall be such that they permit a minimum spacing of 6 inches between any 2 adjacent resistors, and 4 inches between any resistor and the chamber walls.

4.7.15.1.3 Procedure.- A dc potential of 100 volts shall be applied across the end terminals of each resistor whose total resistance is 10,000 ohms or more. Resistors with a total resistance of less than 10,000 ohms shall have sufficient dc potential applied to cause 10 ±1 ma to flow through the entire resistive element. In instances where 100 volts or 10 ma exceed the power rating, rated voltage shall be used (see 4.5.1). The resistive contact-arm load shall be chosen so that the contact-arm current varies from a minimum value to a maximum value of 0.50 ±0.10 ma. The voltage specified above shall be applied intermittently, 1-1/2 hours on and 1/2 hour off, for the duration of the test. The operating shaft shall be continuously cycled through not less than 90 percent of the electrical travel at a rate of 40 ±5 rpm. A cycle shall consist of travel through at least 90 percent of electrical travel and return to the starting point. The total number of cycles to be performed for qualification inspection shall be 500,000 cycles; and for quality conformance inspection, 50,000 cycles. Total resistance shall be measured as specified in 4.7.5.1, in the test chamber at the end of the test, and after the following intervals: Qualification inspection, 50,000 cycles, quality conformance inspection, 10,000 cycles.

4.7.15.2 Load (3.16).-

4.7.15.2.1 Mounting.- Resistors shall be mounted in accordance with 4.7.15.1.1.

4.7.15.2.2 Test conditions.- Test conditions shall be in accordance with 4.7.15.1.2, except that there shall be no circulating air over the resistor other than that created by the resistor. The contact arm shall be positioned at approximately 50 percent of electrical travel.

4.7.15.2.3 Procedure.- The resistor shall be subjected to rated dc voltage (see 4.5.1). In no case, shall the rated voltage be exceeded. This potential shall be applied between the end terminals intermittently, 1-1/2 hours on and 1/2 hour off, for

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a total of 900 ±12 hours. Total resistance shall be measured as specified in 4.7.5.1, in the test chamber at the end of the 1/2-hour-off periods, after 100 ±8, 250 ±12, 500 ±12, 750 ±12, and 900 ±12 hours have elapsed. The resistor shall then be removed from the test chamber and within 24 ±4 hours after removal, the total resistance shall again be measured as specified in 4.7.5.1. The resistor shall also be examined for evidence of mechanical damage and opening of winding.

4.7.16 Low-temperature exposure.- The resistor shall be placed in a cold chamber at room temperature. The temperature shall be decreased to -50° to +0° -5° C within a minimum period of 2 hours. For quality conformance inspection only, at the option of the manufacturer, the resistor may be placed in the cold chamber when the chamber is already at the extreme low temperature. Twenty-four hours after the resistor has reached this temperature, the temperature shall be gradually increased to room temperature within a period of not more than 8 hours. The resistor shall then be removed from the chamber and maintained at room temperature for a period of approximately 24 hours, after which the total resistance shall be measured as specified in 4.7.5.1, and the resistors examined for mechanical damage (see 3.17).

4.7.17 Vibration, high frequency (see 3.18).- The resistor shall be tested in accordance with Method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting of specimens: The resistor shall be mounted by its normal mounting means on a vibration test jig. The contact arm shall be set at approximately 50 percent of electrical travel, and the operating shaft locked (locking device shall prevent rotational movement only and shall not restrict axial movement). The jig shall be so constructed as to preclude any resonances within the test range. The resistor shall be mounted in relation to the test equipment in such manner that the maximum stress applied is perpendicular to a tangential plane through the point of contact of the wiper with the resistance element.
(b) Test condition, as specified in test condition letter A.
(c) Direction of motion: The resistor shall be vibrated in each of two mutually perpendicular planes, one perpendicular and the other parallel to the shaft.
(d) Measurements during vibration: A suitable method shall be provided for the resistor to monitor or register automatically momentary discontinuity of 0.1 millisecond or greater duration, and also to check the continuity between the contact-arm terminal, the tap terminals, and one of the end terminals.
(e) Measurements after vibration: Total resistance between end terminals, and between tap terminals and an end terminal, shall be measured as specified in 4.7.5.1 and 4.7.9, and the resistor shall be examined for mechanical and electrical damage.

4.7.18 Salt spray (corrosion) (see 3.20).- The resistor shall be tested in accordance with Method 101 of MIL-STD-202. The following details shall apply:

(a) Special mounting: On an aluminum panel. When mounted so that the open end of the resistor is exposed, a cover conforming to MS28931 may be used.
(b) Test condition, as specified in test condition letter A.

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(c) Measurements after exposure: The resistor shall be thoroughly washed for 1 minute in free-running tapwater, the temperature of which shall not exceed 38° C. The resistor shall then be placed in an oven maintained at 50° ±3° C, for a period of 24 ±4 hours. At the end of this period, the resistor shall be removed from the oven and examined for corrosion and mechanical operation and tested for electrical continuity.

4.7.19 Humidity cycling (see 3.21).-

4.7.19.1 Mounting.- The resistor shall be mounted by its normal mounting means on an aluminum panel so that the resistor body shall not contact the floor of the test chamber.

4.7.19.2 Procedure.- The resistor shall be placed in a test chamber maintained at a temperature between 20° and 38° C with controlled humidity. During the first 2-hour period, the temperature shall be gradually raised to 71° ±2° C. The temperature of 71° ±2° C shall be maintained during the next 6-hour period. During the next 16-hour period, the temperature in the chamber shall be gradually reduced to between 38° and 20° C. This constitutes one cycle. The relative humidity throughout the cycle shall be 95 ±5 percent. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 120 hours (5 cycles). Distilled or demineralized water having a pH value between 6.5 and 7.5 at 40° C shall be used to obtain the desired humidity. The velocity of the air throughout the test chamber shall not exceed 150 feet per minute. At the end of the last cycle, the resistor shall be removed from the chamber and allowed to dry in free air for a period of 2 hours at room conditions (see 4.5). At the end of the drying period, resistors shall be examined for mechanical damage, and the total resistance shall be measured as specified in 4.7.5.1.

4.7.20 Shock, medium impact (see 3.19).- The resistor shall be tested in accordance with Method 205 of MIL-STD-202. The following details shall apply:

(a) Mounting of specimens: The resistor shall be mounted by its normal mounting means on a test fixture which would permit orientation of the specimen in each of the directions indicated in the paragraph titled "Number and direction of applied shocks". The contact arm shall be set at approximately 50 percent of electrical travel, and the operating shaft locked (locking device shall prevent rotational movement only and shall not restrict axial movement).
(b) Test condition: As specified by test Condition A in the paragraph titled "Height of drop".
(c) Number and direction of applied shocks: As specified in the paragraph titled "Number and direction of applied shocks".
(d) Measurements during shock tests:
(1) Measurements during vibration: A suitable method shall be provided for the resistor to monitor or register automatically momentary discontinuity of 0.1 millisecond or greater duration, and also to check the continuity between the contact-arm terminal, the tap terminals, and one of the end terminals.

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(e) Measurements after shock tests:
(1) Measurements after vibration: Total resistance between end terminals and between tap terminals and an end terminal shall be measured as specified in 4.7.5.1 and 4.7.9, and the resistor shall be examined for mechanical and electrical damage.

4.8 Inspection of preservation, packaging, packing, and marking.- Preservation, packaging, packing, and marking shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Resistors shall be cleaned to comply with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Drying procedure(s) shall not be damaging to the item.

5.1.1.3 Preservative application.- None required.

5.1.1.4 Unit packaging.- The open end of the resistor shall be enclosed by a suitable cover for protection from dust. The resistor shall then be adequately cushioned to prevent damaging movement and individually packaged to comply with submethod IA-8 of MIL-P-116. The packaged item shall then be placed in a close-fitting carton conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packaging.- None required.

5.1.2 Level C.- Resistors shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This package may conform to the supplier's commercial practice for retail distribution when it conforms to the requirements of this level.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.- The packaged item(s) shall be packed in fiberboard containers conforming to PPP-B-636, weather resistant, style optional, special requirement. In lieu of the closure and waterproof requirements of the appendix to PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, ends, edges, and manufacturer's joints with waterproof tape, 3 inches wide, conforming to class 1 of PPP-T-60. Banding reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636. Metal banding shall not be used.

5.2.2 Level B.- The packaged item(s) shall be packed in fiberboard containers conforming to class domestic of PPP-B-636, style and use requirements optional.

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5.2.3 Level C.- The packaged item(s) shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This pack shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when it conforms to the requirements of this level.

5.3 Marking.- In addition to any special marking required by the contract or order, each unit package, intermediate, and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General.- Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical items to the greatest extent possible.

6. NOTES

6.1 Intended use.- The resistor is used in flight simulator systems. Taps at user-selected points give desired relationships between input and output voltages. To eliminate multiple stocking of resistors, identical except for the tap pattern, the resistor is designed to be tapped by the user.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type designation (see 1.2).
(c) Location at which inspection shall be made if other than at supplier's facilities or commercial laboratory (see 4.1).
(d) Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
(e) Cover, when applicable (see 3.22).

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, ATTN: RAAE-5, Navy Depatment, Washington, D. C. 20360.

6.4 Definitions.- For the purpose of this specification, the following definitions apply:

6.4.1 Terminal linearity.- The terminal linearity is the maximum deviation of the actual output from a straight reference line drawn through the zero and total applied voltage points which are separated by the actual effective electrical travel.

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6.4.2 End point.- The end point is the shaft position where the first measurable change of output voltage in excess of 40 percent of resolution is observed as the wiper moves from the overtravel region toward the resistance element.

6.4.3 Tap.- A tap is the connection made to the resistance element at any point.

6.4.4 Gang.- A gang is an assembly of two or more resistors operated by a common drive shaft.

6.4.5 Electrical travel.- Electrical travel is the travel between end points across the resistor element.

6.4.6 Mechanical travel.- Mechanical travel is the travel between the end points of the overtravels on the sides farthest from the resistor element, across the overtravels and the resistor element.

Custodians:
Navy - WP
Air Force - (85)

Reviewer activities:
Navy - WP
Air Force - (85)

Preparing activity:
Navy - WP

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
MIL-R-0903A(ASG) Resistor, Variable, Linear Precision, Tappable

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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